

from the so-called butterfly effects we saw in [Chapter 11](#), actually discovered by Lorenz using weather-forecasting models. Small changes in input, coming from measurement error, can lead to massively divergent projections—and that generously assumes that we have the right equations.

We have polluted for years, causing much damage to the environment, while the scientists currently making these complicated forecasting models were not sticking their necks out and trying to stop us from building these risks (they resemble those “risk experts” in the economic domain who fight the previous war)—these are the scientists now trying to impose the solutions on us. But the skepticism about models that I propose does not lead to the conclusions endorsed by anti-environmentalists and pro-market fundamentalists. Quite the contrary: we need to be hyper-conservationists ecologically, since we do not know what we are harming *with* now. That’s the sound policy under conditions of ignorance and epistemic opacity. To those who say “We have no proof that we are harming nature,” a sound response is “We have no proof that we are not harming nature, either;” the burden of the proof is not on the ecological conservationist, but on someone disrupting an old system. Furthermore we should not “try to correct” the harm done, as we may be creating another problem we do not know much about currently.

One practical solution I have come up with, based on the nonlinearities in the damage (under the assumption that harm increases disproportionately with the quantities released), and using the same mathematical reasoning that led to my opposing the “too big” concept, is to spread the damage across pollutants—should we need to pollute, of course. Let us carry on a thought experiment.

Case 1: You give the patient a dose of cyanide, hemlock, or some poisonous substance, assuming they are equally harmful—and assuming, for the case of this experiment, the absence of super-additivity (that is, no synergetic effects).

Case 2: You give the patient a tenth of a dose of each of ten such substances, for the same total amount of poison.

Clearly we can see that Case 2, by spreading the poison ingested across substances, is at the worst equally harmful (if all the poisonous substances act in the same way), and at the best close to harmless to the patient.

Species Density